

VRMLiveViewer AutoCamera JSON Generation Guide

Unofficial reference for generating and refining camera work from natural language instructions
Document Version: 1.2

This guide summarizes the correspondence between JSON saved by VRM Live Viewer Auto Camera Settings and the UI, together with behavior verified through actual testing. It is intended for generative AI that creates and refines AutoCamera JSON.

1. Basic JSON structure

```
{ "dataType": "AutoCamera", "mode": "Follow", "param": { "keys": [...], "values": [...], "datas": [ { "keys": [...], "values": [...], "etc": { "keys": [], "values": [] }, "version": "3.14.1" } ] }
```

Important: within each object, keys[n] corresponds to values[n]. Preserve order and element counts.

2. Frames and common settings

UI / Function	JSON key	Observed mapping
Start Frame	startFrame	Start frame; frame-based and affected by fps
Switching Mode	switchingMode	Frame observed
FPS Unit	fps	30 / 60 verified; affects startFrame, freezeFrame, etc.
Far Clip Plane	farClip	UI Clip Distance; verified minimum 150. Unit, maximum and real-distance mapping not established
Max Chara No	maxCharaIndex	UI 1 → 0; (All) → -1
Chara No	charaIndex	UI 1 → 0
Each Setting / Multi Chara	eachSetting / multiChara	eachSetting: False=shared, True=individual; multiChara: False=single, True=multiple

3. Basic camera settings

UI / Function	JSON key	Observed mapping
Offset X/Y/Z	offset	X,Y,Z
Distance	distance	Base camera distance
Rotation X/Y/Z	angle	X,Y,Z
VAngle	viewAngle	View angle
Fix Position	fixed	"" = no fixed axes; "X,Y,Z" observed
Position	position	Camera position
Speed	speed	Movement speed when following target

4. Animated/change settings

UI / Function	JSON key	Observed mapping
Changing Position	enablePositionChange	True/False
Position Interpolation	positionInterp	Interpolation string
One Way (Position)	positionHalf	True=one-way; False=back-and-forth
Add X/Y/Z	positionMax	X,Y,Z
Changing Time / Offset	positionTime / positionTimingShift	Time-setting value / start shift
Changing Rotation	enableRotationChange	True/False
Rotation Mode	rotationMode	Add / MinMax. Mode selection exists only for Rotation
Rotation Interpolation	rotationInterp	Interpolation string
One Way (Rotation)	rotationHalf	True=one-way; False=back-and-forth
Min / Max Angle	rotationMin / rotationMax	MinMax: start→end + target rotation. Add: 0→added angle + target rotation
Changing Time / Offset	rotationTime / rotationTimingShift	rotationTime=24 verified as 12 sec at 30/60fps
Changing Distance	enableDistanceChange	True/False
Distance Interpolation	distanceInterp	Interpolation string
One Way / Add Distance	distanceHalf / distanceMax	True=one-way; False=back-and-forth. distanceMax is additive movement from base distance
Changing VAngle	enableVAngleChange	True/False
VAngle Interpolation	vangleInterp	Interpolation string
One Way / Max VAngle	vangleHalf / vangleMax	True=one-way; False=back-and-forth / stored value

5. Jitter

UI / Function	JSON key	Observed mapping
Jitter enabled	enableJitter	True/False
Position Amount / Freq	jitterPosAmount / jitterPosFreq	Position shake amount / frequency
Position Axis X/Y/Z	jitterPosAxisRate	X,Y,Z
Rotation Amount / Freq	jitterRotAmount / jitterRotFreq	Rotation shake amount / frequency
Rotation Axis X/Y/Z	jitterRotAxisRate	X,Y,Z

6. Interpolation

UI	Saved JSON	Motion guide
Linear	Linear	Linear / constant
Ease-In-Out	EaseInOut	Gradually faster → gradually slower

UI	Saved JSON	Motion guide
Ease-In-Out (low)	EaseInOutLow	Observed as milder / more Linear-like
Ease-In	EaseIn	Gradually faster
Ease-In (low)	EaseInLow	Observed as milder / more Linear-like
Ease-Out	EaseOut	Gradually slower
Ease-Out (low)	EaseOutLow	Observed as milder / more Linear-like

Note: The author's exact internal handling/strength of the (low) interpolation variants remains unverified. However, EaseInOutLow, EaseInLow, and EaseOutLow are valid values observed in saved JSON. A JSON-generating AI does not need to account for the internal interpolation strength and may use them like the other interpolation choices.

7. Combining camera effects

Position / Rotation / Distance / VAngle / Jitter may be combined within one camera-data entry. Do not assume "1 camera = 1 effect." When moving a multi-entry part, preserve its internal startFrame offsets.

new startFrame = requested recipe start + (original startFrame - first startFrame of the part)

8. Continuous rotations of 360° or more

A 360° rotation can be specified in one data entry. Use rotationHalf=True for one-way motion. Chaining multiple 0→360 entries produced a slight reverse movement at boundaries, so prefer a cumulative angle in one entry for continuous rotation. 0→720 (2 turns) and 0→3600 (10 turns) were verified. Do not generalize beyond 3600 as unlimited.

enableRotationChange=True / rotationMode=MinMax / rotationHalf=True / rotationMin=0,0,0 / rotationMax=0,360,0 / rotationTime=24 / rotationTimingShift=0

rotationTime=24 was verified as 12 seconds at both 30fps and 60fps. It is not the same frame-based time system as startFrame / freezeFrame. Do not convert 12 seconds to rotationTime=360. A camera starting at S for N frames occupies S through S+N-1; the next camera starts at S+N.

9. Natural-language camera terms

Term	Meaning in this guide
Horizontal 360° orbit	Character-centered horizontal full orbit
Roll	Rotate the screen around the viewing axis
Dutch angle	Static tilted composition
90° roll / sideways composition	Character appears to lie sideways
Dolly / truck	Camera position movement using Changing Position
Jitter	Handheld-like or intense shake

10. Refinement guidance

Request	UI	JSON
Tilt too strong	Rotation Z	angle Z
Zoom too strong	Distance / Add Distance	distance / distanceMax
Rotation speed	Changing Rotation - Changing Time	rotationTime
Shake too strong	Jitter Amount	jitterPosAmount / jitterRotAmount
Sideways move too large	Changing Position - Add X	positionMax X

11. Output rules

- Preserve keys[n] and values[n] correspondence, order, and element counts.
- **Strictly follow the sample JSON structure. In every object, keys[n] and values[n] must correspond one-to-one. Keep the keys and values arrays in the same order and with the same element count. Vector values such as X,Y,Z must remain a single string, as in the sample, and must not be split across multiple values. Before output, verify that keys and values have equal element counts in every datas entry.**
- Output the final VRM Live Viewer file as one-line minified JSON.
- Pretty-printed JSON may be used for explanation, but also provide the minified final file.
- Do not change stored strings such as True/False into different tynes

12. Rules for generative AI

- Prefer structures and values from a close example in the supplied sample JSON.
- Combine Position / Rotation / Distance / VAngle / Jitter in one camera where appropriate.
- Do not use a fixed camera-count rule. Choose density from song length, BPM, intensity and camera duration, and avoid long nearly-static stretches.
- As a general rule, place camera switches on musical bar boundaries. Do not restrict switching strictly to bar boundaries when another musically natural point exists, such as immediately after a rest.
- If adjacent cameras would have almost the same composition, do not switch using only tiny differences in Position, Rotation, Distance, or VAngle. Either create a clear compositional difference or retain the previous composition so the cut does not appear as an unintended small jump.
- Prefer cumulative angles for continuous rotations of 360° or more.
- Preserve fps, total frames, character conditions and BPM.
- Do not invent unverified meanings, units or keys.
- Do not infer or invent song sections, scenes, or directing intent that the user did not provide, and do not use such assumptions as a basis for camera type, placement, or switching timing. Generate camera work from the user's supplied conditions/instructions, this guide, and the sample JSON.

Pending: Appropriate Distance, Position, Offset, VAngle, and related ranges depend on the number of characters.
No universal numeric range for keeping characters in frame is defined at this time; a single-character range also remains pending testing.

13. Example JSON

samples/full_camera_sample_6660f_160bpm_v1.2.json is an example output containing multiple camera effects and compound effects.
It is not a preset for a particular song. Conditions: 160 BPM, 30 fps, start frame 0, total length 6660 frames.

Camera	Start Frame	Effect
01	0	Horizontal 360° rotation + distance change
02	360	Position move + distance change + 20° Dutch angle
03	600	Roll 0°→90° + view-angle change
04	840	Position move + distance change + jitter
05	1080	Horizontal rotation −90°→90° + 45° Dutch angle + distance change
06	1320	Vertical position move + view-angle change + jitter
07	1560	Wide horizontal rotation −120°→120° + distance change
08	1800	Continuous 720° horizontal rotation (2 turns) + distance change
09	2160	Position move + distance change + view-angle change + −30° Dutch angle
10	2400	Distance change + view-angle change (→83.53706°) + jitter
11	2640	Position move + roll 0°→−90° + −45° Dutch angle
12	2880	Wide horizontal rotation −120°→120° + distance change + jitter
13	3120	60° pitch + vertical position move + distance change
14	3360	Roll 0°→−90° + view-angle change + 30° Dutch angle
15	3600	Horizontal 360° rotation + distance change
16	3960	Position move + roll 0°→−90° + distance change
17	4200	Large distance change + view-angle change + jitter
18	4440	Roll 0°→180° + 45° Dutch angle + distance change
19	4680	Roll 0°→−90° + distance change + jitter
20	4920	Position move + horizontal rotation + view-angle change
21	5160	Position move + distance change + 45° Dutch angle
22	5400	Horizontal 360° rotation + position move
23	5760	Distance change + view-angle change + jitter
24	6000	Roll 0°→180° + position move + distance change
25	6240	Position move + view-angle change + 45° Dutch angle + jitter
26	6480	Horizontal rotation −90°→90° + distance change
27	6600	Final static composition (−20° Dutch angle)

Unofficial status

This generation guide is unofficial documentation created by MoonaWorks. It is not an official specification from the developer of VRM Live Viewer. Please do not send questions about this project to the developer of VRM Live Viewer.